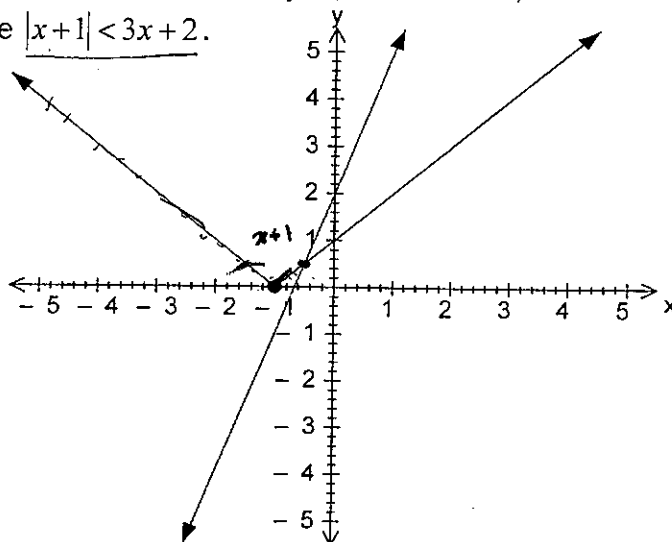


Question 1 (14 Marks)

Marks

- (a) Solve $x^2 - 6x + 3 = 0$ by completing the square. Give your answer in surd form. 3
- (b) Solve simultaneously $y - 2x = 1$ and $x^2 + y^2 = 10$ 3
- (c) Solve
- (i) $x^2 \geq 16$ 1
- (ii) $|2x - 1| < 3$ 2
- (iii) $\frac{x+3}{x-3} \geq 3$ 3
- (d) The graph below shows the functions $y = |x+1|$ and $y = 3x+2$. 2

Using the graph to assist you, or otherwise,
solve $|x+1| < 3x+2$.



Question 2 (16 Marks) START ON A NEW PAGE **Marks**

- (a) Consider the function $f(x) = \sqrt{x+1}$. 6
- (i) State the domain and range of $f(x)$
- (ii) Sketch the function
- (iii) Find the inverse $f^{-1}(x)$
- (b) Sketch the graph of $y = -(x-2)^3$ showing its key features. 2
- (c) Show whether the $f(x) = \frac{2x}{3(1+x^2)}$ is an odd function, an even function or neither. 2
- (d) Consider the function $f(x) = \frac{2x}{x^2 + 2x - 3}$. 6
- (i) Find the x and y intercepts
- (ii) Find the equations of any vertical asymptotes
- (iii) Find the equation of the horizontal asymptote
- (iv) Sketch the graph of the function.

Question 3 (23 Marks) START ON A NEW PAGE

- (a) Find the exact value of $\cot 120^\circ$ 1
- * (b) If $\tan \theta = 2$ and $\sin \theta < 0$, find the exact value of $\cos \theta$. 2
- (c) Solve the following equations for $0^\circ \leq \theta \leq 360^\circ$ 7
- (i) $2 \sin \theta = -\sqrt{3}$
- (ii) $3 \cos^2 \theta + 2 \cos \theta = 0$ (nearest minute)
- * (iii) $\sec \frac{\theta}{2} = -2$
- (d) Eliminate θ from the pair of equations to find a relationship between x and y: 2
- $x = 1 + \sin \theta$
- $y = 1 + \cos \theta$
- (e) Show that $\frac{\cot \theta}{\operatorname{cosec} \theta - 1} + \frac{\cot \theta}{\operatorname{cosec} \theta + 1} = 2 \sec \theta$ 3

Q3 continued on next page

Question 3 continued

Marks

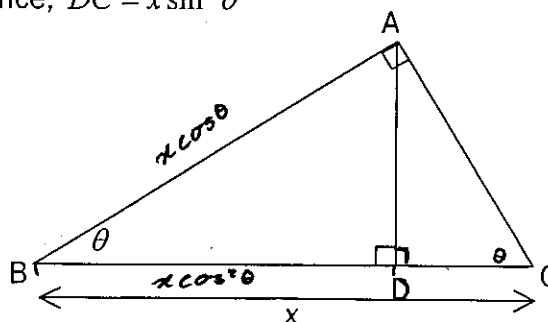
* (f) $\triangle ABC$ and $\triangle ABD$ are right-angled. Show that

4

(i) $AB = x \cos \theta$

(ii) $BD = x \cos^2 \theta$

(iii) Hence, $DC = x \sin^2 \theta$

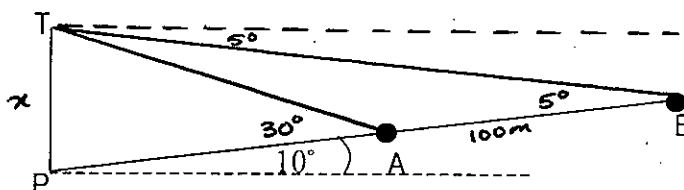


(g) P, A, and B are points on a ramp which makes an angle of 10° to the horizontal as shown on the diagram below. From A and B, the angles of elevation of the top T of a flagpole at P were 30° and 5° respectively. The distance AB is 100 metres.

4

(i) Copy and complete the diagram below.

(ii) Calculate the height of the tower to the nearest centimetre.



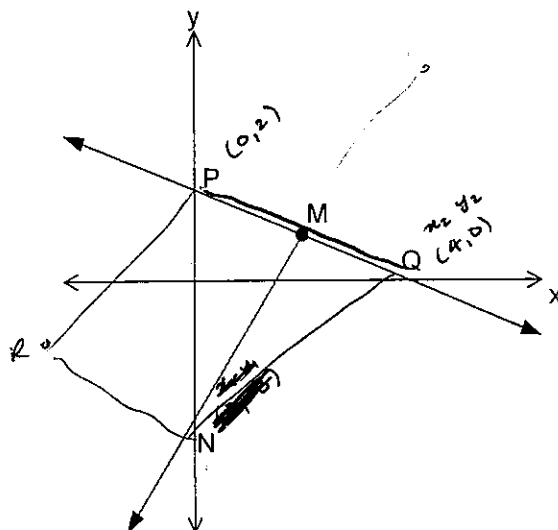
Q4 on next page

Question 4 (16 marks) START ON A NEW PAGE

Marks

- (a) Find the coordinates of the point P that divides the interval AB, where A is $(-3, 5)$ and B is $(-6, -10)$, externally in the ratio 2:3. 3

(b)



The diagram above shows the points $P(0,2)$ and $Q(4,0)$. The point M is the midpoint of PQ . The line MN is perpendicular to PQ and meets the y -axis at N .

- | | | |
|--------|--|---|
| (i) | Show that the gradient of PQ is $-\frac{1}{2}$. | 1 |
| (ii) | Find the coordinates of M. | 1 |
| (iii) | Find the equation of the line MN. | 2 |
| (iv) | Show that N has coordinates (0,-3). | 1 |
| (v) | Find the distance NQ. | 1 |
| (vi) | Find the equation of the circle with centre N and radius NQ. | 2 |
| (vii) | Show that this circle passes through the point P. | 1 |
| (viii) | Find the coordinates of the point R such that PRQN is a rhombus. | 1 |

- (c) The perpendicular distance between the point $(a, -2)$ and the line $3y = 4x + 2$ is 8 units. Find any possible values of a .

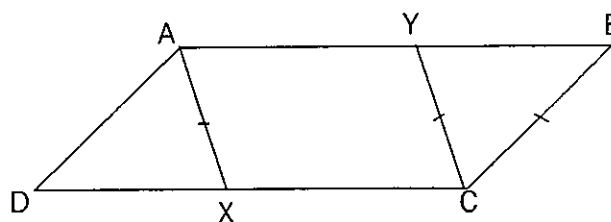
Q5 on next page

Question 5 (16 Marks) START ON A NEW PAGE

Marks

(a)

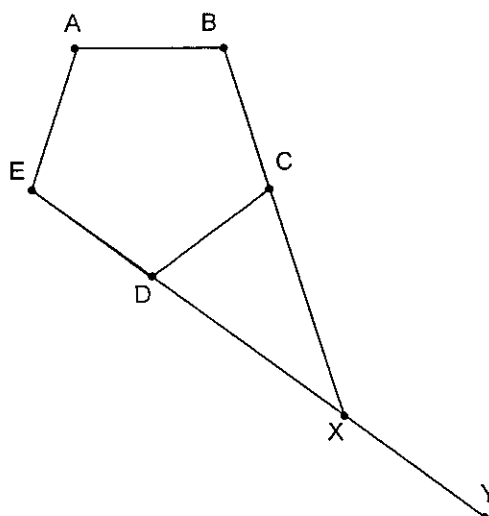
6



ABCD is a parallelogram. The point X lies on CD and the point Y lies on AB. $AX = XC = BC$ as shown on the diagram.

- (i) Explain why $\hat{AD}X = \hat{CB}Y$.
- (ii) Show that $AD = AX$
- (iii) Show that triangles ADX and CBY are congruent.
- (iv) Hence prove that AYCX is a parallelogram.

(b)



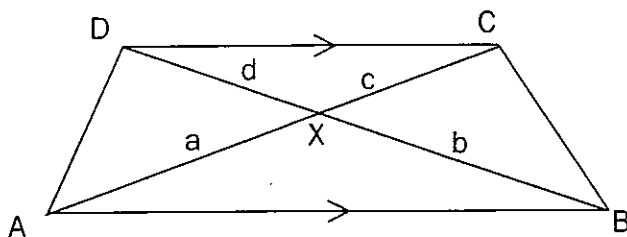
In the diagram ABCDE is a regular pentagon and BC and ED produced meet at X. The point Y lies on EDX produced. 3

- (i) Find the size of $\hat{BCD}$.
- (ii) Find the size of $\hat{CXY}$ giving reasons.

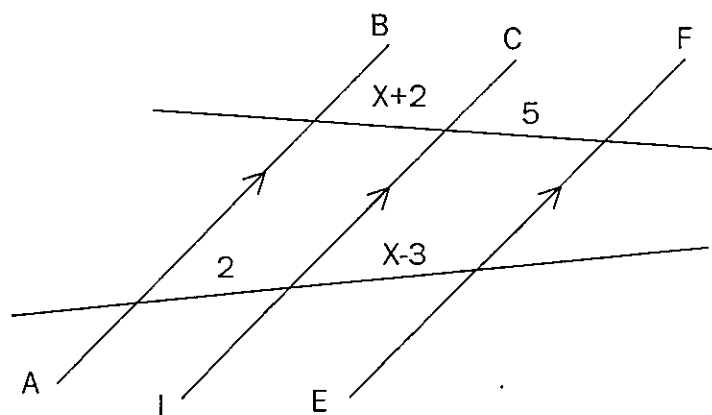
Q5 continued on next page

Question 5 continued

Marks

(c) 4

- (i) Prove that $\triangle DXC \cong \triangle BXA$.
 (ii) Hence, prove that $\triangle DAX$ and $\triangle CBX$ have the same area.

(d) 3

Given that $AB \parallel CD \parallel EF$, find the value of x giving reasons.

END OF PAPER

Q1

a) $x^2 - 6x = -3$
 $x^2 - 6x + \left(\frac{-6}{2}\right)^2 = -3 + \left(\frac{-6}{2}\right)^2$
 $x^2 - 6x + 9 = -3 + 9$
 $(x-3)^2 = 6$
 $x-3 = \pm\sqrt{6}$
 $x = 3 \pm \sqrt{6}$

b) $x^2 + y^2 = 10$ ①
 $y - 2x = 1$ ②
 From ② $y = 2x + 1$
 sub $y = 2x + 1$ in ①
 $x^2 + (2x+1)^2 = 10$
 $x^2 + 4x^2 + 4x + 1 = 10$
 $5x^2 + 4x - 9 = 0$
 $(5x+9)(x-1) = 0$
 $x = -9/5$ or 1

$x = -9/5$ $y = 2(-9/5) + 1$
 $= -13/5$

$(-9/5, -13/5)$

$x = 1$ $y = 3$
 $(1, 3)$

c) i) $x^2 \geq 16$

$x^2 - 16 \geq 0$

$(x-4)(x+4) \geq 0$

$x \leq -4$ or $x \geq 4$

ii) $|2x-1| < 3$
 $\pm(2x-1) < 3$
 $2x-1 < 3$ $-(2x-1) < 3$
 $2x < 4$ $-2x+1 < 3$
 $x < 2$ $-2x < 2$
 $x > -1$
 $-1 < x < 2$

iii) $\frac{x+3}{x-3} \geq 3$ $x \neq 3$
 $x(x-3)^2 \times (x-3)^2$

$(x+3)(x-3) \geq 3(x-3)^2$

$(x+3)(x-3) - 3(x-3)^2 \geq 0$

$(x-3)(x+3-3(x-3)) \geq 0$

$(x-3)(-2x+12) \geq 0$

$3 \leq x \leq 6$
 but $x \neq 3$

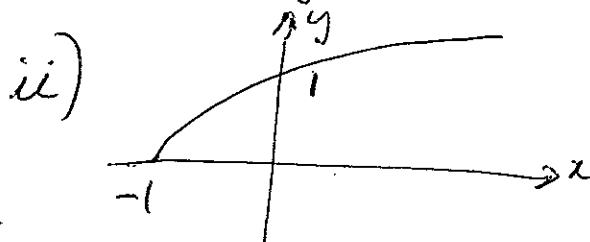
$\therefore 3 < x \leq 6$

d) $y = x+1$ ① (positive branch of $y = |x+1|$)
 $y = 3x+2$ ②
 $① = ②$
 $0 = -2x-1$
 $x = -1/2$

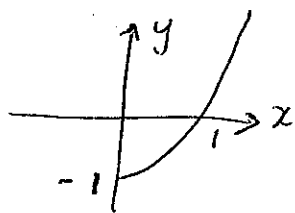
$\therefore x > -1/2$

Q2

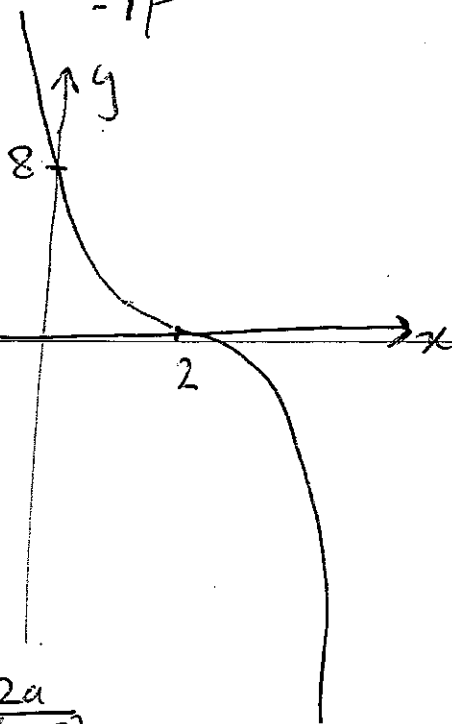
- a) i) domain $x \geq -1$
~~range~~ range $y \geq 0$



- iii) $x = \sqrt{y+1}$
 $x^2 = y+1$
 $y = x^2 - 1$
 for $x \geq 0$



b)



c) $f(a) = \frac{2a}{3(1+a^2)}$

$$f(-a) = \frac{2(-a)}{3(1+(-a)^2)} = \frac{-2a}{3(1+a^2)} = -f(a)$$

$\therefore$ ODD

d) $f(x) = \frac{2x}{x^2+2x-3}$
 $= \frac{2x}{(x+3)(x-1)}$

- i) $x=0$ $y=0$ $(0,0)$
 x & y intercept is the origin

- ii) $x = -3$ and $x = 1$

iii) $\lim_{x \rightarrow \infty} \frac{2x}{x^2+2x-3}$

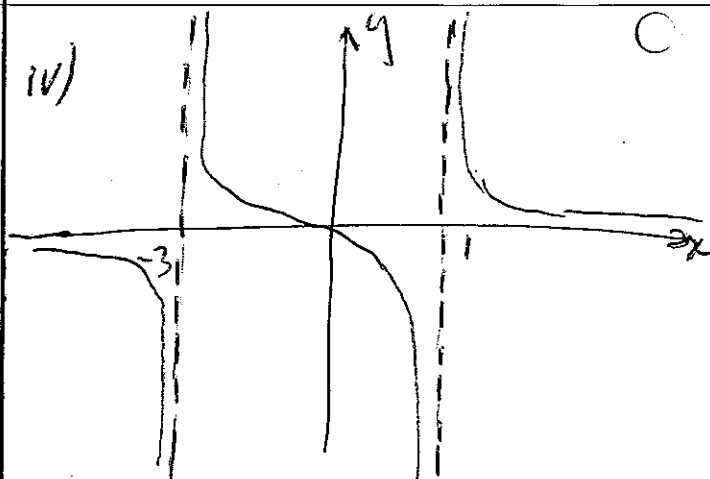
$$= \lim_{x \rightarrow \infty} \frac{\frac{2x}{x^2}}{\frac{x^2}{x^2} + \frac{2x}{x^2} - \frac{3}{x^2}}$$

$$= \lim_{x \rightarrow \infty} \frac{\frac{2}{x}}{1 + \frac{2}{x} - \frac{3}{x^2}}$$

$$= \frac{0}{1} = 0$$

$\therefore y = 0$

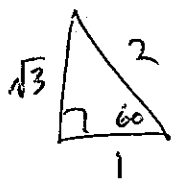
iv)



x	-4	-1	1/2	2
y	-16	0.5	0.2	0.8

Q3 a) $\cot 120 = \frac{-1}{\tan 60}$

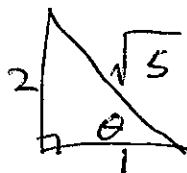
$$= \frac{-1}{\sqrt{3}}$$



b) $\tan \theta = 2$

$\sin \theta < 2$

$\cos \theta = \frac{1}{\sqrt{5}}$



c) i) $\sin \theta = -\sqrt{3}/2$

$\theta = 180 + 60, 360 - 60$
 $= 240, 300^\circ$

ii) $3 \cos^2 \theta + 2 \cos \theta = 0$

$\cos \theta (3 \cos \theta + 2) = 0$

$\cos \theta = 0$ or $\cos \theta = -\frac{2}{3}$

$\theta = 90^\circ$ or 270° $\theta = 180 - 48^\circ$
 or $180 + 48^\circ$

$\therefore \theta = 90^\circ, 270^\circ, 131^\circ 49', 228^\circ 11'$

iii) $\sec \frac{\theta}{2} = -2$

$\cos \frac{\theta}{2} = -\frac{1}{2}$

$\frac{\theta}{2} = 180 - 60, 180 + 60$

$= 120, 240$

$\theta = 240$ or 480

But $0 \leq \theta \leq 360$

$\therefore \theta = 240^\circ$

d) $x = 1 + \sin \theta$ ①

$y = 1 + \cos \theta$ ②

From ① $\sin \theta = x - 1$

From ② $\cos \theta = y - 1$

$\sin^2 \theta + \cos^2 \theta = 1$

$\therefore (x-1)^2 + (y-1)^2 = 1$

e) LHS = $\frac{\cot \theta}{\operatorname{cosec} \theta - 1} + \frac{\cot \theta}{\operatorname{cosec} \theta + 1}$

$= \frac{\cot \theta (\operatorname{cosec} \theta + 1) + \cot \theta (\operatorname{cosec} \theta - 1)}{\operatorname{cosec}^2 \theta - 1}$

$= \frac{2 \cot \theta \operatorname{cosec} \theta}{\operatorname{cosec}^2 \theta - 1}$

From $\sin^2 \theta + \cos^2 \theta = 1$
 $1 + \cot^2 \theta = \operatorname{cosec}^2 \theta$
 $\cot^2 \theta = \operatorname{cosec}^2 \theta - 1$

$= \frac{2 \cot \theta \operatorname{cosec} \theta}{\cot^2 \theta}$

$= \frac{2 \operatorname{cosec} \theta}{\cot \theta}$

$= \frac{2}{\sin \theta} \times \frac{\sin \theta}{\cos \theta}$

$= \frac{2}{\cos \theta}$

$= 2 \sec \theta = \text{RHS.}$

Q 3 continued

$$f) i) \cos \theta = \frac{AB}{x}$$

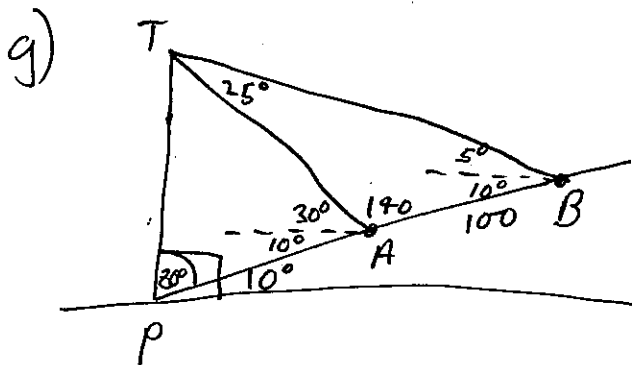
$$\therefore AB = x \cos \theta$$

$$ii) \cos \theta = \frac{BD}{AB}$$

$$\cos \theta = \frac{BD}{x \cos \theta}$$

$$BD = x \cos^2 \theta$$

$$\begin{aligned} iii) BC &= BC - BD \\ &= x - x \cos^2 \theta \\ &= x(1 - \cos^2 \theta) \\ &= x \sin^2 \theta \end{aligned}$$

In $\triangle ATB$

$$\frac{100}{\sin 25^\circ} = \frac{AT}{\sin 15^\circ}$$

$$AT = \frac{100 \sin 15}{\sin 25}$$

In $\triangle APT$

$$\frac{AT}{\sin 80} = \frac{TP}{\sin 40}$$

$$TP = \frac{\sin 40 \left(\frac{100 \sin 15}{\sin 25} \right)}{\sin 80}$$

$$= 39.97 \text{ m}$$

Q 4

$$a) \begin{matrix} (-3, 5) & (-6, -10) \\ & \swarrow \searrow \\ & -2:3 \end{matrix}$$

$$\left(\frac{3x-3+(-2)x-6}{-2+3}, \frac{3x5+(-2)x(-10)}{-2+3} \right)$$

$$= \frac{-9+12}{1}, \frac{15+20}{1}$$

$$= (3, 35)$$

$$\begin{aligned} b) i) m &= \frac{y_2 - y_1}{x_2 - x_1} \\ &= \frac{0 - 2}{4 - 0} \\ &= \frac{-2}{4} \\ &= -\frac{1}{2} \end{aligned}$$

$$\begin{aligned} ii) \text{Midpt} &= \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right) \\ &= \left(\frac{0 + 4}{2}, \frac{2 + 0}{2} \right) \\ &= (2, 1) \end{aligned}$$

$$iii) m = \frac{-1}{-\frac{1}{2}} = 2 \quad (2, 1)$$

$$y - y_1 = m(x - x_1)$$

$$y - 1 = 2(x - 2)$$

$$y = 2x - 4 + 1$$

$$y = 2x - 3$$

$$iv) x = 0 \quad y = 2(0) - 3 = -3$$

$$\therefore N \text{ is } (0, -3)$$

$$v) (0, -3) (4, 0)$$

$$D = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$= \sqrt{(0 - 4)^2 + (-3 - 0)^2}$$

$$= \sqrt{16 + 9} = 5$$

Q4 continued

$$b) (x-0)^2 + (y+3)^2 = 25$$

vi)

$$vii) x=0$$

$$(0-0)^2 + (y+3)^2 = 25$$

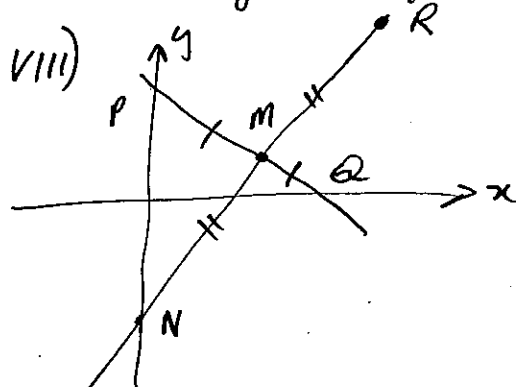
$$(y+3)^2 = 25$$

$$y+3 = \pm\sqrt{25}$$

$$y = -3 \pm 5$$

$$= 2 \text{ or } -8$$

$P(0, 2) \therefore$ circle passes through the point P



M is the mid point of NR if $PRQN$ is a rhombus.

Let R be (x, y)

$$\text{Then } (2, 1) = \left(\frac{x+0}{2}, \frac{y+(-3)}{2} \right)$$

$$\frac{x}{2} = 2$$

$$x = 4$$

$$\frac{y-3}{2} = 1$$

$$y-3 = 2$$

$$y = 5$$

$\therefore R$ is $(4, 5)$

$$c) 0 = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

$$4x - 3y + 2 = 0$$

$$0 = 8 \quad (a, -2)$$

$$8 = \frac{|4a + -3(-2) + 2|}{\sqrt{4^2 + (-3)^2}}$$

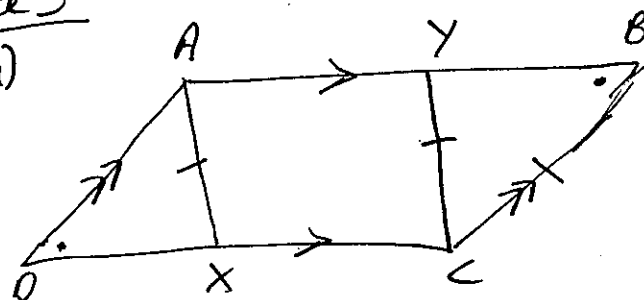
$$8 = \frac{|4a + 8|}{5}$$

$$40 = |4a + 8|$$

$$\begin{array}{l|l} 4a + 8 = 40 & -(4a + 8) = 40 \\ 4a = 32 & -4a - 8 = 40 \\ a = 8 & -4a = 48 \\ & a = -6 \end{array}$$

Q5

a)



i) $\angle ADX = \angle CBY$
(opposite angles in a parallelogram)

ii) $AD = BC$ (opposite sides of a parallelogram)

$BC = AX$ (given)

$\therefore AD = AX$

iii) In $\Delta s ADX$ and CBY

$\angle ADX = \angle YBC$ (above)

$\angle ADX = \angle AXD$ (base angles of an isosceles Δ)

Q5 continued

$\angle CYB = \angle CBY$ (base angles of an isosceles Δ)

$$\therefore \angle AXD = \angle BYC$$

$$AX = CY \text{ (given)}$$

$$\therefore \Delta ADX \equiv \Delta CBY \text{ (AAS)}$$

iv) $DX = BY$ (corresponding sides of congruent Δ s)

$$AB = DC \text{ (opposite sides of a parallelogram)}$$

$$XC = DC - DX$$

$$AY = AB - BY$$

$$\therefore XC = AY$$

$$AX = YC \text{ given}$$

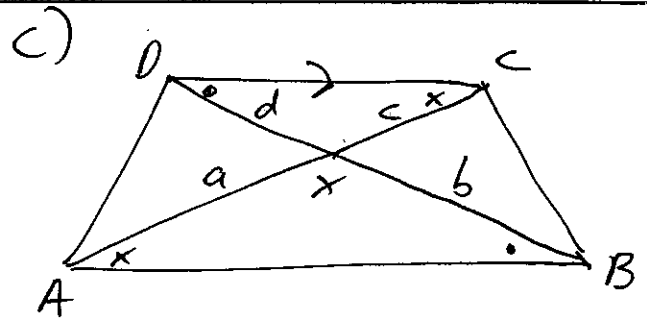
$\therefore AXCX$ is a parallelogram (opposite sides equal)

$$\begin{aligned} \text{b) } \angle BCD &= \frac{180 \times (5-2)}{5} \\ &= 108^\circ \end{aligned}$$

$$\begin{aligned} \angle XCD &= 180 - 108 \\ &= 72^\circ \text{ (supplementary } \angle\text{s)} \end{aligned}$$

$$\text{Similarly } \angle CDX = 72^\circ$$

$$\therefore \angle CXY = 72 + 72^\circ = 144^\circ \text{ (exterior angle of a triangle)}$$



i) In Δ s DXC and BXA

$$\angle CDX = \angle XBA \text{ (alternate } \angle\text{s } DC \parallel AB)$$

$$\angle DCX = \angle XAB \text{ (alternate } \angle\text{s } DC \parallel AB)$$

$$\therefore \Delta DXC \equiv \Delta BXA \text{ (equiangular)}$$

ii) $\frac{b}{d} = \frac{a}{c}$ (corresponding sides in same ratio)

$$\therefore bc = ad$$

$$\text{Area } \Delta DAX = \frac{1}{2} ad \sin \angle AXD$$

$$\Delta CXB = \frac{1}{2} bc \sin \angle CXB$$

$$\angle CXB = \angle AXD \text{ (vert. opp)}$$

$\therefore$ areas are equal.

$$\text{d) } \frac{2}{x-3} = \frac{x+2}{5}$$

$$10 = (x-3)(x+2)$$

$$10 = x^2 - x - 6$$

$$0 = x^2 - x - 16$$

$$x = \frac{-(-1) \pm \sqrt{1^2 - 4 \times (-16)}}{2}$$

$$= \frac{1 \pm \sqrt{1+64}}{2}$$

$$= \frac{1 \pm \sqrt{65}}{2}$$

(parallel lines cut intercepts in the same ratio)